2.5 Homework: Mental Math Solutions

Students write only the answer. The notes give the step a student should be doing in their head, and the mistakes worth looking for.

1. Fill in the blanks.

   (a) 1.6 km = **1600 m**
   (b) 85 cm = **0.85 m**
   (c) 0.25 m = **25 cm**
   (d) 4 mm = **0.4 cm**
   (e) 2.2 m = **2200 mm**
   (f) 500 mm = **0.5 m**

   Going to a smaller unit makes the number bigger. For (d), 1 cm = 10 mm, so 4 mm is four tenths of a centimeter.

2. Squares and square roots.

   (a) $3^2 = \mathbf{9}$
   (b) $7^2 = \mathbf{49}$
   (c) $12^2 = \mathbf{144}$
   (d) $20^2 = \mathbf{400}$
   (e) $0.5^2 = \mathbf{0.25}$
   (f) $0.3^2 = \mathbf{0.09}$
   (g) $\sqrt{81} = \mathbf{9}$
   (h) $\sqrt{144} = \mathbf{12}$

   The trap is (f): 0.3^2 *is* $0.3 \times 0.3 = 0.09$*, not 0.9. Squaring a number less than 1 makes it smaller. Square roots undo squares, which is the last step when solving* $y = \frac{1}{2}gt^2$ *for* t*.*

3. $\frac{1}{2} \times a \times t^2$.

   (a) $\frac{1}{2} \times 10 \times 4 = \mathbf{20}$
   (b) $\frac{1}{2} \times 10 \times 9 = \mathbf{45}$
   (c) $\frac{1}{2} \times 4 \times 25 = \mathbf{50}$
   (d) $\frac{1}{2} \times 0.2 \times 100 = \mathbf{10}$

   Only t *is squared. A student who squares the whole product, or computes* $(\frac{1}{2} \times 10 \times 2)^2 = 100$ *for (a), has the order of operations wrong.*

4. Speed units.

   (a) 10 m/s = **36 km/h**
   (b) 20 m/s = **72 km/h**
   (c) 108 km/h = **30 m/s**
   (d) 360 km/h = **100 m/s**

   From m/s to km/h, multiply by 3.6; back again, divide. 108 is 3×36*, and 360 is* 100×3.6*. The number in km/h is always the bigger one.*

5. Circle the best answer.

   (a) **(A) 0.3.** *1 ft = 0.3048 m exactly. 3.28 is the number of feet in a meter, and 12 is the inches in a foot.*

   (b) **(B) 1.6.** *1 mi* $\approx$ *1.609 km. 0.6 is the reverse, miles in a kilometer.*

   (c) **(B) 27.** *60 mi/h* $\approx$ *97 km/h, which is close to 100 km/h, and* $100 \div 3.6 \approx 28$*. The exact value, 26.8 m/s, was given in 2.3. Choice (D) is the speed in km/h.*

6. Fractions as decimals.

   (a) $\frac{2}{5} = \mathbf{0.4}$ (c) $\frac{3}{8} = \mathbf{0.375}$ (e) $\frac{1}{20} = \mathbf{0.05}$ (g) $\frac{1}{6} = \mathbf{0.1\overline{6}}$

   (b) $\frac{4}{5} = \mathbf{0.8}$ (d) $\frac{7}{8} = \mathbf{0.875}$ (f) $\frac{1}{25} = \mathbf{0.04}$ (h) $\frac{5}{6} = \mathbf{0.8\overline{3}}$

   Each fifth is 0.2 and each eighth is 0.125. $\frac{1}{20}$ *is half of* $\frac{1}{10}$*;* $\frac{1}{25}$ *is like 4 cents in a dollar.* $\frac{1}{6}$ *is half of* $\frac{1}{3}$*. Accept* $0.1666\ldots$ *and* $0.8333\ldots$*, or 0.17 and 0.83 written with* $\approx$*.*

7. Fractions of an hour and a day.

   (a) 0.25 hour = **15 min**

   (b) 0.3 hour = **18 min**

   (c) 1.5 min = **90 s**

   (d) 150 s = **2.5 min**

   (e) 12 min = **0.2 h**

   (f) 20 min = $\mathbf{\frac{1}{3}}$ **h**

   (g) 1 day = **1440 min**

   (h) 1.75 hours = **1 h 45 min**

   Each tenth of an hour is 6 minutes. For (g), $24 \times 60 = 1440$*. As in 2.1, part (h) is the one to watch: 0.75 h is 45 min, not 75 min.*

8. Class time.

   (a) **1.5 h** $2 \times 45 = 90$ min, which is $1\frac{1}{2}$ h.

   (b) **54 h** $36 \times 1.5 = 36 + 18 = 54$.

9. **About 6 miles.** $10 \times \frac{5}{8} = \frac{50}{8} = 6\frac{1}{4}$.
   Accept 6 to 6.25 mi; the true value is 6.21 mi.

10. Falling stone with $g \approx 10$ m/s^2.

    (a)

t (s)	1	2	3	4	5
speed (m/s)	**10**	**20**	**30**	**40**	**50**
distance fallen (m)	**5**	**20**	**45**	**80**	**125**

    The speed grows in equal steps; the distance grows as t^2*, so it is* $5 \times 1, 4, 9, 16, 25$*. With* $g = 9.80$ *m/s*^{2} *the distances are 4.90, 19.6, 44.1 m, as in the classwork.*

    (b) **8 s.** $5t^2 = 320$, so $t^2 = 64$ and $t = 8$.

    (c) **Between 8 and 9 s.** $5t^2 = 381$ gives $t^2 \approx 76$, between $8^2 = 64$ and $9^2 = 81$.
    With $g = 9.80$ *m/s*^{2} *the time is 8.82 s. Air resistance would make the real fall slower; this is Giancoli's King Kong problem.*